🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.DS.6C

Mean Absolute Deviation

Lesson 8-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the mean absolute deviation (MAD) to describe how spread out data is.

Language Objective: I can explain my work using the words mean absolute deviation, deviation, absolute value, and spread.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Mean Absolute Deviation | **Deviation** | **Absolute Value** | **Spread** | **Data distribution** | **Variability**

👆 *Tap any word to see what it means and a picture.*

1. The average distance of each value from the mean is the ________.
2. How far a single value is from the mean is its ________.
3. The distance of a number from zero, always positive, is its ________.
4. How far apart the data values are from each other is the ________.
5. The way data values are spread out is the ________.
6. How much the data values differ is the ________.

Reflect — Exit Ticket

Data set: 4, 8, 6, 10, 2. The mean is 6. What is the MAD?

A. 2.4

B. 6

C. 0

D. 12

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Mean Absolute Deviation
2. **Notes 2:** Deviation
3. **Notes 3:** Absolute Value
4. **Notes 4:** Spread
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Mean Absolute Deviation is skipping the key idea: "MAD = the average of how far each number is from the mean (always a positive distance)." — always check your work against this rule before you submit.*
8. **Try It 1:** B. 2 — *Add the distances: 3 + 1 + 1 + 3 = 8. Divide by the 4 games: 8 ÷ 4 = 2. The MAD is 2 points.*
9. **Try It 2:** A. Shooter A, because a smaller MAD means scores stay closer to the average — *A smaller MAD means the scores hug the mean. Shooter A's MAD of 2 is smaller than Shooter B's 6, so Shooter A is more consistent.*
10. **Exit Ticket:** A. 2.4 — *Deviations: −2, 2, 0, 4, −4. Absolute deviations: 2, 2, 0, 4, 4. Sum = 12. MAD = 12 ÷ 5 = 2.4.*